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Total No. of printed pages = 03

Winter, 2025

B.Tech 2nd Semester Examination

Programming for Problem Solving – Theory

Course Code: BCS23101T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

1. Choose the correct option (10 x 1 mark = 10 marks)

a) Which of the following is a logical error in a program?

- i) Missing semicolon at the end of a statement
- ii) Using incorrect syntax of a programming language
- iii) Using a variable without declaring it
- iv) Using a wrong formula that gives incorrect results but no error during compilation

b) Which of the following best describes an algorithm?

- i) A pictorial representation of a program
- ii) A set of instructions designed to perform a specific task
- iii) A compiled version of a computer program
- iv) A hardware component used for computation

c) Which of the following is a valid identifier in C?

- i) Int
- ii) 2value
- iii) _sum1
- iv) Float

d) What will be the output of the following C code?

```
int x = 10, y = 20;
if (x > y)
    printf("X is greater");
else
    printf("Y is greater");
```

- i) X is greater
- ii) Y is greater
- iii) Error: Invalid syntax

iv) No output

e) How many times will the inner loop execute in the following C code?

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 2; j++) {  
        printf("Loop\n");  
    }  
}
```

- i) 2 ii) 5 iii) 6 iv) 4
- f) Given the matrices A and B below, what will be the value at position [1][1] of matrix C after $C = A + B$?

```
A = [ [1, 2],  
      [3, 4]]  
B = [ [5, 6],  
      [7, 8]]
```

- i) 10 ii) 11 iii) 12 iv) 13
- g) What will be the output of the following C code?

```
char str1[20] = "Hello";  
char str2[20] = "World";  
strcat(str1, str2, 2);  
printf("%s", str1);
```

- i) Hello ii) World iii) HelloWorld iv) HelloWo

h) Which keyword is used in C to define a structure?

- i) struct
ii) structure
iii) record
iv) object

i) Which of the following correctly defines a function with no return type and no arguments?

- i) void func() { printf("Hello"); }
ii) void func { printf("Hello"); }
iii) int func() { printf("Hello"); }
iv) void func() { return; }

j) What will be the output of the following C program?

```
#include <stdio.h>  
void displayMessage() {  
    printf("Hello, World!\n");  
}  
int main() {
```



```
    displayMessage();  
    return 0;  
}
```

- i). Hello, World!
- ii). Error: Function must return a value
- iii). No output
- iv). Warning: No return statement in displayMessage

2. Write an algorithm and draw a flowchart to: 5+5
- (a) Swap two numbers.
 - (b) Greatest of three numbers.

or

Draw the diagram and explain the compilation and execution process of a C program. 10

3. Write a program
- (a) To find the sum of even numbers between 20 to 70. 5
 - (b) To display the multiplication table of 11. 5

Or

Write a program using switch case. Take inputs of two numbers. Press 1 to perform addition, press 2 to perform subtraction, press 3 to perform multiplication, press 4 to perform division. If any number is pressed display message "Wrong option". 10

4. Answer the following.
- (a) Write a program to find the largest number in an array of 9 numbers. 6
 - (b) Provide the syntax for declaring a two-dimensional array. Write about different ways to initialize a 2D array. 4

Or

Write a C program to store and display the records of B.Tech 2nd year with 60 students. 10

5. Answer the following.
- (a) What is a function? Write the advantages of using a function. 5
 - (b) Write a program to find the sum of two numbers using function. 5

Or

Write a program using function. Take as input marks of three subjects. Find the average. If average is between 80-100, print "first division", between 60-79, print "second division", between 40-59, print "third division" else print "fail". 10